



Master Thesis Proposal

How do you feel today?
Multimodal Emotion Recognition: Predicting Human Emotions in
Automotive Environments based on Speech and Facial Expressions

Author: Deniz Cevher
Supervisors: Sebastian Zepf
Roman Klinger

Institut für Maschinelle Sprachverarbeitung
University of Stuttgart
70569 Stuttgart, Germany
deniz.cevher@ims.uni-stuttgart.de
roman.klinger@ims.uni-stuttgart.de

Advanced Engineering - RD/UI
Daimler AG
059-X832 D-71059, Sindelfingen
deniz.cevher@daimler.com
sebastian.zepf@daimler.com

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1 Introduction

A major challenge in the automotive industry is to constantly increase safety for drivers, passengers, and everybody involved in motor traffic. While mechanical malfunctions in vehicles may cause some accidents, a major cause of accidents is the lack of drivers' attention (Beanland et al., 2013). Drivers tend to be more distracted by happy music and show a more calm and steady driving when listening to sad music for example (Pêcher et al., 2009). Galovski and Blanchard (2004) state that frustrated drivers tend to show aggressive behaviour and can be accounted for a high number of accidents. Also, angry drivers were shown to be more prone to errors as stated by Jeon (2015). Thus, a promising approach to increasing safety is to understand the driver's emotional mental state and build user interfaces that can help in increasing a driver's attention or decrease a driver's frustration level. The very first step in this direction is to be able to consistently and reliably detect the driver's emotional state.

Over a century ago James (1884) posed the question of what emotions are. This question is still not answered completely, but we know that emotions are an important part of human life. For example, they play an important role in communication, help in making decisions and even facilitate learning (Poria et al., 2017a). Cabanac (2002) summarizes multiple definitions and comes to the conclusion that emotions can be defined as mental states. A lot of researchers have attempted to describe these mental states using many different indicators like facial expressions (e.g., Ekman et al., 1969), neural activity (e.g., Lane et al., 1997), physiological reactions like changes in heart rate or increase in adrenaline (e.g., H. Funkenstein, 1955) and linguistic cues in prosody (e.g., W. Frick, 1985), syntax (e.g., Alm et al., 2005), and in semantics (e.g., Tokuhisa et al., 2008). Notwithstanding the ongoing debate about whether the human ability to have emotional mental states is innate or learned, humans have a precise intuition of how to interpret other humans' emotions. Children as young as 3 are already able to identify basic emotions in facial expressions and body poses (Parker et al., 2013), as well as in the content and prosody of an utterance (Morton and Trehub, 2001).

With the emergence of affective computing in the last 20 years, research on emotions is no longer restricted to the fields of biology, psychology, or social science, but has now attracted the attention of fields such as computer science, natural language processing and artificial intelligence. By recognizing emotion and sentiment, affective computing is attempting to enable computational systems to simulate empathy by predicting and mimicking human emotions. While sentiment analysis aims at recognizing affect from data on a coarse-grained level, emotion recognition (ER) does so on a more fine-grained level by classifying data into a greater range of classes (Poria et al., 2017a). Such data may be of different modalities, like recorded speech, texts, videos, and photos. Many researchers have attempted to use data from *one* modality to recognize emotions. A problem is that a unimodal (UM) ER system relies solely on the presence of input from that specific modality and cannot process data from any other modality. This may be a major disadvantage in (potential) real-world applications such as adaptive user interfaces (UI) (e.g., Meudt et al., 2016), where users do not always express their emotions through the same communicative channel. Unimodal systems can also be disadvantageous in situation in which information from a certain modality is distorted or impossible to record, such as an adaptive

in-car UI. For instance, relying only on visual information might cause problems at night where facial expressions may not be recognized very well due to insufficient video quality when there is only little light in the car. Likewise, it may be hard to rely solely on linguistic data for it may be absent (driving alone), sparse or not context related. A multimodal ER system integrated into an in-car UI would be able to more accurately predict emotion in these unfavorable circumstances and additionally improve the accuracy of emotion detection when information from all modalities is available at the same time (Poria et al., 2017b).

To resolve the issues of unimodal systems research towards multimodal (MM) ER is being done recently. While most attempts combine two modalities to predict human emotions, some even include a third modality (compare D’Mello and Kory, 2014). Poria et al. (2017b) and Rozgić et al. (2012), for example, use visual data in combination with textual information and audio features to build multimodal ER systems.

Given the hypothesis that multimodal systems outperform unimodal systems, this research will focus on the in-car application of multimodal ER. To the best of our knowledge, there has been no attempt to combine (1) *linguistic features* like syntactic structures and semantic meaning, (2) *paralinguistic features* such as the melody and tone of an utterance and (3) *visual features* as provided by facial expressions, in an in-car environment. This is the main contribution of this work.

2 Related Work

Text based emotion recognition Recognizing discrete emotions from textual data is a task that is being addressed by many researchers. Alm et al. (2005) recognize passages in fairy tales which contain emotionally filled content and classify these passages into two classes (positive or negative). They use frequency features, emotion lexicons and morphological cues as features for flat classification. Other approaches predict the mood of blog posts using Support Vector Machines (SVM) (Mishne, 2005). Mostly the feature sets are hand crafted and the results rely heavily on the choice of appropriate and useful features. To overcome this issue Su et al. (2018) use word embeddings and affective lexicons to extract semantic and emotional word vectors from the segmented text. In that study the recognition accuracy of a long-short term memory (LSTM) using these features was above 70%.

Audio based emotion recognition Studies on audio based emotion recognition mainly utilize a feature set consisting mostly of mel frequency cepstral coefficients (MFCC), log-Mel filterbank and prosodic features extracted from spoken utterances (Dellaert et al., 1996; Kachele et al., 2014). These features can be extracted easily and have a small number of dimensions. This makes these features very suitable for low-resource domains. Recently most studies use neural networks like LSTMs (Chernykh et al., 2017) and convolutional neural networks (CNN) (Neumann and Vu, 2017).

Multimodal fusion In multimodal ER information from different modalities is fused or combined. There are two main approaches to fusing multiple modal-

ities: (1) Early fusion or feature-level fusion and (2) late fusion or decision-level fusion. These two approaches differ in the timing of the fusion. (1) Feature-level fusion concatenates features extracted from different modalities to one general feature vector. Poria et al. (2015) fused text, audio and video features extracted from the eINTERFACE corpus and were able to achieve a significant increase in accuracy at the time. (2) Dobrišek et al. (2013) used decision-level fusion to combine the matching scores of two separate systems (one for ER from audio and the other for ER from video) to obtain a combined matching score.

Domain adaption and transfer learning Generally, the approach to building a model includes training and test data that stems from the same distribution or domain. Unfortunately, this means that for every new domain there needs to be labeled data, which is very costly and involves a lot of manual annotation. Domain shifting is a promising technique to work around this issue. In domain shifting the training and test sets are taken from different distributions. Ben-David et al. (2007), for example, train a part-of-speech tagger on data from the financial domain and evaluate the tagger with data from the biomedical domain. They introduce a measure to determine the distance between two domains which can be used to find suited source domains. Blitzer et al. (2007) tackle the task of sentiment analysis and apply domain shifting, using Amazon review for four different types of products. Oquab et al. (2014) approach the problem of limited target data using transfer learning. They pre-train CNN layers using images from the ImageNet dataset (source task). Their target task is to classify objects and actions from the Pascal VOC. Therefore they transfer all but the last layer of the source model to the target model and add adaption layers to it. The adaption layers are then trained (fine-tuned) using the limited data from the Pascal VOC.

3 Goals

In this research, we will investigate the influences that different modalities have in a multimodal emotion recognition system. The main hypothesis is that data from different modalities contain different information, i.e. the data is complementary and thus the performance of a multimodal ER system increases as compared to a unimodal ER system. We will develop an emotion classifier that uses features from three different modalities, namely (1) *linguistic features* from text, (2) *audio features* from audio, (3) *video features* from videos.

The main goal of this research is to *determine which of the three chosen modalities (text, audio, video) to combine in order to achieve the best performance for a multimodal in-car ER system.*

Since there is no corpus available for in-car emotion data, we will first collect new data in a driving simulator study, thus, creating a new corpus. This newly created corpus (and subsets of it) will serve as an evaluation corpus throughout all experiments.

Due to limited resources this corpus will likely be too small to be used as a training set. Thus, another objective is to identify for the text modality a suited domain (corpus) which can be used for training an emotion recognizer for in-car utterances. We will thus find domains/corpora that suite this purpose best.

Finally we will investigate the best fusion method for the three modalities, comparing feature-level fusion and decision-level fusion. This includes building separate models for each modality.

Type	Training set	Test set
Domain Adaption	$C_i \in K$	M
Domain Mixing	$C_i \in K \cup C_j \in K$	M
Transfer Learning	$C_i \in K$ (pre-train) and	$M_e \subset M$
	$M_t \subset M$ (fine-tune)	$M_t \cap M_e = \emptyset$

Table 1: Experiments to be conducted in the text phase. C_i is one random corpus contained in the corpus collection K . The same holds for C_j . M is the manually collected corpus.

4 Work Plan

The work plan is subdivided into two main parts. First, we will focus on data collection as described in section 4.1. This includes creating a multimodal corpus in a driving simulator study as well as acquiring existing corpora. After the data acquisition phase, we will use the data in deep learning experiments. These experiments are described in section 4.2.

4.1 Data

Creation of a Multimodal Corpus We will collect and annotate data in a driving simulator study and use this data mainly for evaluation (let this corpus be the manually collected corpus). For training we will use existing corpora and parts of the manually collected corpus.

The driving simulator study will be split into two phases: The emotion elicitation phase and the emotion annotation phase.

(1) In the *emotion elicitation phase*, participants will be asked to drive in a virtual city (including highways) and reach a specific location in that city as quickly as possible (using some hints from a simulated navigation system), while obeying all traffic rules. During the ride other simulated drivers will perform ordinary manouvers in the close vicinity of the participant’s verhicle. These manouvers (emotion elicitation events) should elicit the participant’s emotions. After each emotion elicitation event (EEE) the participant is asked to describe the situation she experienced on the phone (we will use a Wizard of Oz setup to simulate phone calls). Additionally, the participants will be asked simple and open interview questions, which they can freely answer. During this phase, we record the participant’s answers, descriptions and facial expressions.

(2) After they reached the goal the ride is over and participants are asked to take a seat in front of a screen for the *emotion annotation phase*. In this phase, participants will review (on a screen) the EEEs (including the descriptions they gave in the phone calls) and the answers they gave to the interview questions. For each clip they will be asked to pick from a predefined list of emotions, the emotion they felt or perceived most dominantly at the time of the EEE or interview question. Throughout all experimental settings we will use this

manually collected corpus (or a subset of it) as the evaluation set. Let this corpus be M , for manually collected corpus.

Text Data Bostan and Klinger (2018) compiled a list of text corpora, suited for emotion classification. They were able to put twelve different corpora in one uniform format in order to make them easily usable in future studies. Additionally, they translated the annotation schemas used in different corpora into *one* scheme. This collection contains corpora from different domains (such as news, fairytales, etc). This is particularly useful for our purpose since we want to find a domain that is best suited for emotion classification of in-car utterances. We will translate all corpora into German using a machine translation method. For clarity, let this collection be K and let $C \in K$ be one corpus from that collection.

Audio Data We will use the Interactive Emotional Dyadic Motion Capture (IEMOCAP) database (Busso et al., 2008) to train an audio classifier. This database consists of recordings taken from improvised speech and scripted scenes. We are especially interested in the audio data provided by this corpus. It contains 10039 samples with an average length of 4.5s. From the annotation provided by the IEMOCAP database we will use only the discrete classes.

Video Data For facial recognition we will use a pre-trained classifier built by a commercial provider.

4.2 Experiments

We will aim at identifying emotion in human utterances and facial expressions. This will be done using a neural network that fuses features and models from three different modalities. The process can be divided into three preliminary steps, where we will choose pre-trained or build own baseline models for each of the three modalities (text, audio, video) first, and a final step where we will identify the best fusion method and investigate the impact each modality has on the system.

Text The first step is to find those text corpora that are best suited as training sets for an emotion classifier for in-car utterances. For this purpose, we will train a deep learning model to identify emotions in transcribed utterances and evaluate its performance using data from the in-car domain. We will perform three different types of experiments in this step. The experiments differ in the way the domains of the training sets are combined and in the way of training the model. Table 1 summarizes the experiments conducted.

(1) Domain Adaption: In the first type of experiment we will perform domain adaption, i.e. we will use from the set of corpora K one corpus C for training and the in-car corpus M for evaluation. The corpus C may be any of the corpora contained in K .

(2) Domain Mixing: As training sets for the second type of experiment we will mix domains by combining two corpora. For twelve corpora (cf. Section 4.1) this results in a total of 66 possible combinations. In order to reduce this number we will use the top five best performing corpora from the domain adaption

experiment described previously. We will thus conduct ten experiments of this type.

(3) Transfer Learning: Thirdly, we will use transfer learning. This type of experiment differs from the other two in that the training includes data from a non-in-car domain as well as parts of the in-car dataset. Unlike in the domain mixing experiments, there will be no direct union of the two datasets. We will rather train different parts of the model with different datasets. Namely, we will pre-train the model using a source corpus C taken from K . By fine-tuning (post-training) the last layer using a subset of the in-car corpus M we will shift the domain of the model towards the target domain (in-car). The textual data will be transformed into word embeddings and used as input to the model.

This procedure will give insights in what kind of corpus and what kind of model is best suited for predicting the emotions of in-car utterances from textual features.

Audio The experimental setup in the audio step will include the IEMOCAP database which we will split into a training set (70%), a development set (20%) and an evaluation set (10%). We will use basic acoustic features, like mfcc and logmel features, as the input to our audio model.

Video We will use a pre-trained model (fixed weights, thus not trainable) for facial recognition. The model predicts seven emotions and 20 facial expressions or action units. We will concatenate these predictions to form a vector, which we will utilize as input to a trainable model.

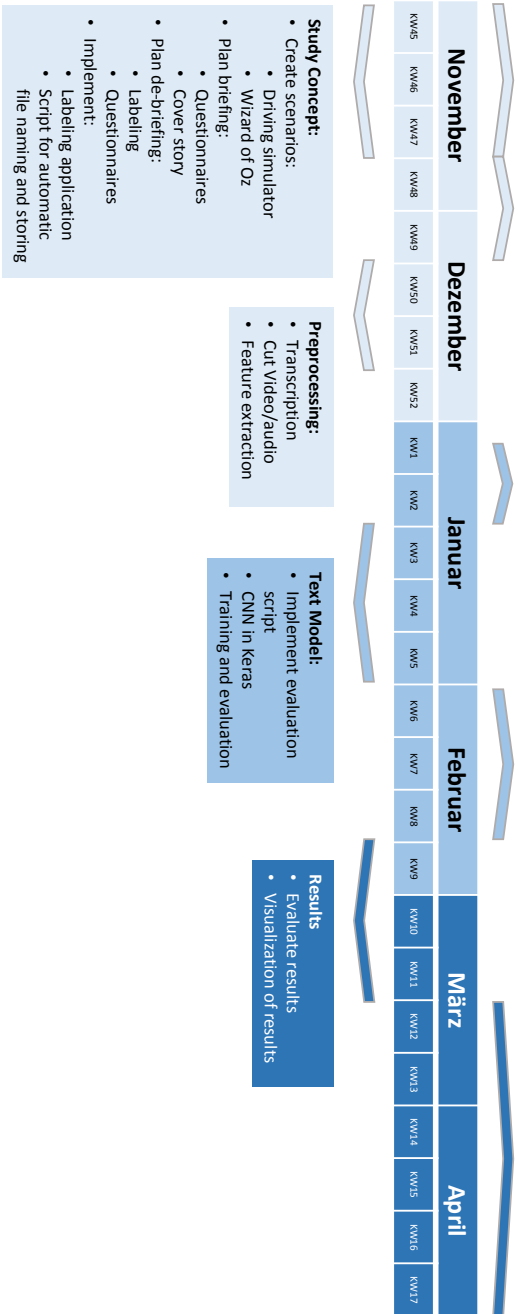
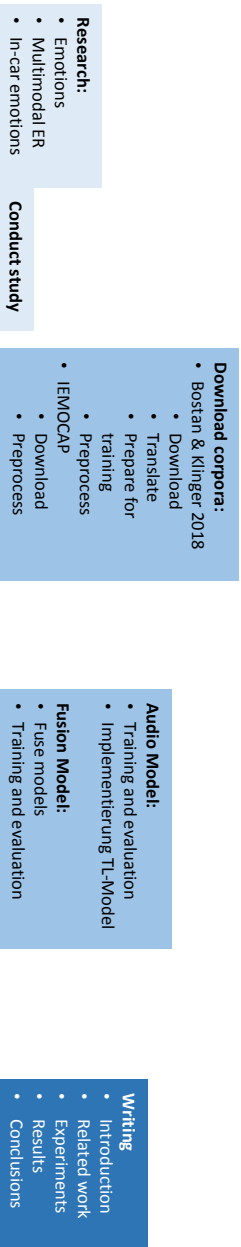
Multimodal Fusion In terms of multimodal fusion, we will first determine the best fusion method. Including all modalities, we will compare feature-level fusion and decision-level fusion. The better of the two methods will be used in the final experimental setup, described below.

We will conduct three final experiments to investigate how the three modalities (text, audio, video) interact with the each other.

For training we will use the corpora described above: The best corpus from the text experiments, the IEMOCAP corpus for audio and the pre-trained facial recognition model. Additionally, we will fine-tune the model using parts of the manually collected corpus.

For evaluation we will use the rest of the manually collected corpus in three experiments. In order to determine the effects of the three modalities, we will mask one modality in each experiments. The masking method depends on the chosen fusion method. In the case of decision-level fusion, the model will ignore the prediction of one modality and make the final prediction based solely on the individual predictions of the remaining two modalities. In case of feature-level fusion, we will use a random or uniform feature vector for the masked modality. This setup results in a total of three experiments: TA_v , TaV and tAV ; where T/t, A/a and V/v represent the text, audio and video modality, respectively. The modality in lower case is the one being masked.

The results can then be compared to the baseline results and to the performance that was achieved using all modalities to find out the individual impact of each modality.



5 Schedule

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